Python Introduction

Python is a high-level, general-purpose language with readable syntax.

1) What is Python, and where is it used?

- **High-level:** you work with ideas like lists and dictionaries, not memory.
- **Interpreted:** the Python runtime executes your code and reports errors.
- **Dynamically typed:** a name can refer to different types at different times.
- **Multi-paradigm:** procedural, object-oriented and functional styles.

Your code `.py` file or cell → Python runtime runs line by line → Output or an error message

Area	Example
Automation	Rename files or generate reports
Web development	Build APIs and server-side applications
Data analysis / AI-ML	Clean datasets, experiment with models
Testing / Education	Automate checks, teach programming concepts

2) Run Python in VS Code

1. Install Python 3 and the extension → 2. Select the interpreter → 3. Save a `.py` file, Run Python File → 4. Notebook: Shift + Enter

3) Your first program

Example :

```
print("Hello, Python!")
print("I am learning programming.")
print(2 + 3)
```

Output :

```
Hello, Python!
I am learning programming.
5
```

text in quotes is printed as written; 2 + 3 is calculated first, so it prints 5

Quick check

Which symbol starts a comment? **#**. What does = do? It **assigns** a value to a name.

VARIABLES & DATA TYPES

Indentation groups code, variables store values, and every value has a type.

1) Comments, indentation and a colon

Example :

```
# This line is a comment
student_name = "Asha"
if student_name:
    print(f"Hello, {student_name}!")
```

Output :

```
Hello, Asha!
```

a hash starts a comment, a colon opens a block, so indent it four spaces

2) Variables and assignment

Example :

```
student_name = "Asha"
age = 19
temperature = 28.5
is_enrolled = True
print(student_name, age, temperature, is_enrolled)
```

Output :

```
Asha 19 28.5 True
```

assignment gives a name a value, it does not test equality. Names are case-sensitive

3) The basic types and type()

Type	Example	Typical use
int	42	whole numbers
float	3.14	decimal numbers
str	"Python"	text
bool	True	logical conditions
NoneType	None	no value yet

Example :

```
price = 49.50
language = "Python"
is_ready = False
print(type(price))
print(type(language))
print(type(is_ready))
```

Output :

```
<class 'float'>
<class 'str'>
<class 'bool'>
```

dynamic typing: the same name can later point to a different type

Identifiers & Operators

An identifier is a name you choose; a keyword is a word Python reserves for itself.

1) Identifier rules

Name	Valid?	Why
`total_marks`	yes	letters + underscore
`_count`	yes	may start with _
`student2`	yes	digit is not first
`2score`	no	starts with a digit
`student-name`	no	hyphen not allowed
`class`	no	keyword: SyntaxError

2) Operators (a = 7, b = 2)

Meaning	Example	Result
addition	`a + b`	9
subtraction	`a - b`	5
multiplication	`a * b`	14
division	`a / b`	3.5
floor division	`a // b`	3
remainder	`a % b`	1
power	`a ** b`	49

Type	Operators	Example	Result
Comparison	`== != < <= > >=`	`a > b`	`True`
Logical	`and or not`	`a > b and b > 0`	`True`

3) All together

Example :

```
a, b = 7, 2
print("Division:", a / b)
print("Floor division:", a // b)
print("Remainder:", a % b)
print("Power:", a ** b)
print("Equal?", a == b)
```

Output :

```
Division: 3.5
Floor division: 3
Remainder: 1
Power: 49
Equal? False
```

Predict first

`print(7 / 2, 7 // 2, 7 % 2)` prints **3.5 3 1**.



Name It Well, Operate With Confidence.

Conversion, Input & Practice

`input()` always returns text. Convert it with `int()` or `float()` before doing maths.

1) Type conversion

Call	What it does	Example	Result
`int()`	text to whole number	`int("19")`	`19`
`float()`	text to decimal	`float("1.72")`	`1.72`
`str()`	value to text	`str(19)`	`'19'`

2) Safe input with f-strings

Example :

```
name = input("Your name: ").strip()
age_text = input("Your age: ").strip()
try:
    age = int(age_text)
except ValueError:
    print("Please enter your age as a whole number.")
else:
    print(f"Hello, {name}!")
    print(f"Next year you will be {age + 1}.")
```

Output :

```
Your name: Asha
Your age: 19
Hello, Asha!
Next year you will be 20.
```

3) Practice tasks

Task 1 Book title, price, availability.

Solution :

```
title = "Python Basics"
price = 349.50
in_stock = True
print(title, price, in_stock)
```

```
Python Basics 349.5 True
```

Task 2 Add two numbers.

Solution :

```
a = float(input("First number: "))
b = float(input("Second number: "))
print("Sum:", a + b)
```

```
First number: 4
Second number: 5.5
Sum: 9.5
```

Task 3 Celsius to Fahrenheit.

Solution :

```
c = float(input("Celsius: "))
f = c * 9 / 5 + 32
print(f"{c} C = {f} F")
```

```
Celsius: 37
37.0 C = 98.6 F
```

Task 4 Predict the output.

Solution :

```
print(7 / 2, 7 // 2, 7 % 2)
```

```
3.5 3 1
```

Quick check

Valid names: **first_name, score2**. Invalid: `first-name`, `2nd_place`, `class`.